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Experiment 7: Configure a local Kubernetes cluster using Minikube and deploy an Nginx Pod using a YAML file.

Kubectl: is command line tool for Kubernetes where we can directly interact with Kubernetes

Minikube: is a local Kubernetes cluster

Before that start the docker desktop engine

Step 1: Minikube installation

Go to browser type minikube installation

<https://minikube.sigs.k8s.io/docs/start/?arch=%2Fwindows%2Fx86-64%2Fstable%2F.exe+download>

To install the latest minikube **stable** release on **x86-64 Windows** using **.exe download**:

1. Download and run the installer for the [latest release](#).

Or if using PowerShell, use this command:

Click on latest release

Cmd->run as administrator

>minikube

>minikube start --driver=docker

>minikube status

If you get the error for the above commands

minikube stop

minikube delete

minikube start --driver=docker

OR

minikube delete

docker version

docker info

minikube start --driver=docker

Step 2: Installation of Kubectl

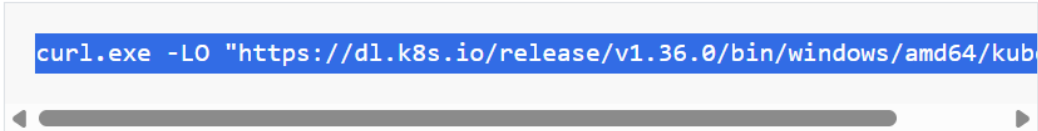
Type kubectl installation-->click on install tools-->choose platform

<https://kubernetes.io/docs/tasks/tools/install-kubectl-windows/>

Copy the Curl and paste in the cmd

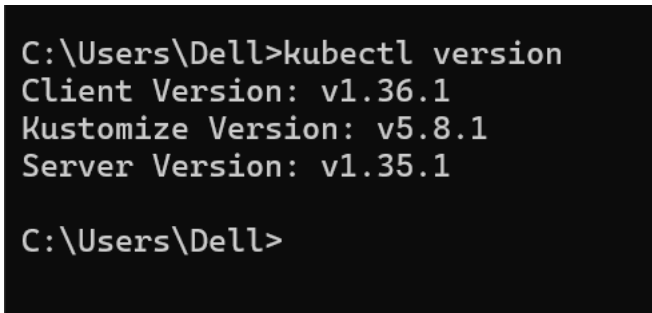
- Using curl:

If you have `curl` installed, use this command:



```
curl.exe -LO "https://dl.k8s.io/release/v1.36.0/bin/windows/amd64/kub
```

>kubectl version

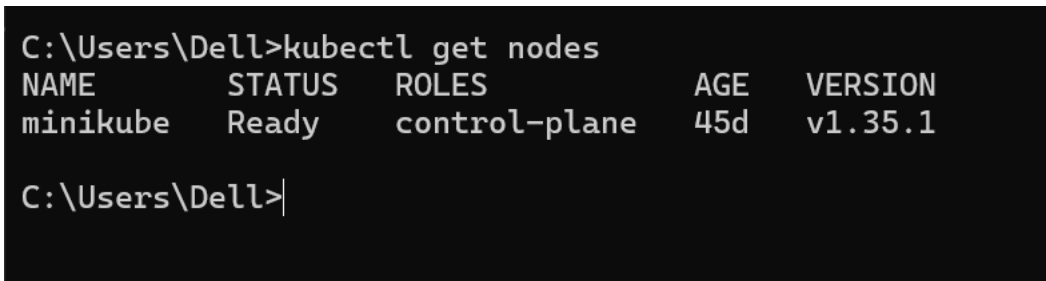


```
C:\Users\Dell>kubectl version
Client Version: v1.36.1
Kustomize Version: v5.8.1
Server Version: v1.35.1

C:\Users\Dell>
```

Check whether kubectl is connected with minikube by the command

>kubectl get nodes



```
C:\Users\Dell>kubectl get nodes
NAME        STATUS    ROLES    AGE   VERSION
minikube    Ready     control-plane  45d   v1.35.1

C:\Users\Dell>
```

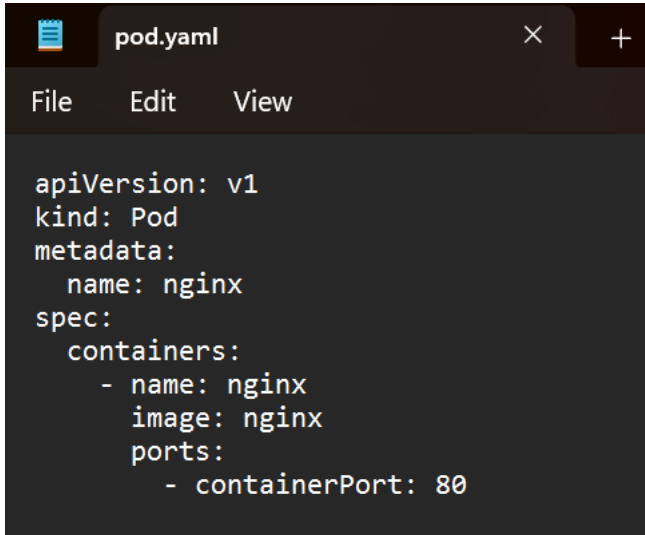
Step 3: Installation of POD

Goto web browser and type Kubernetes pod

<https://kubernetes.io/docs/concepts/workloads/pods/>

Pod is basically a Yaml file

Copy the yaml code



```
apiVersion: v1
kind: Pod
metadata:
  name: nginx
spec:
  containers:
  - name: nginx
    image: nginx
    ports:
    - containerPort: 80
```

Cmd>vi pod.yml

Paste the code (if vi is not recognized)

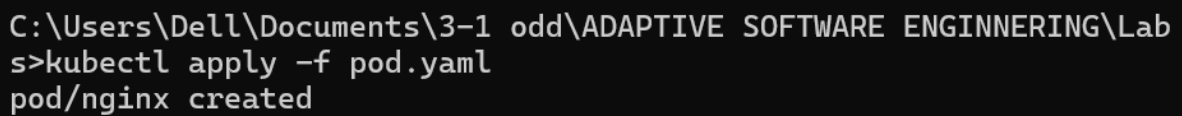
Cmd> notepad pod.yaml

Click yes

Copy the code and Save the file and close Notepad.

Deploy it:

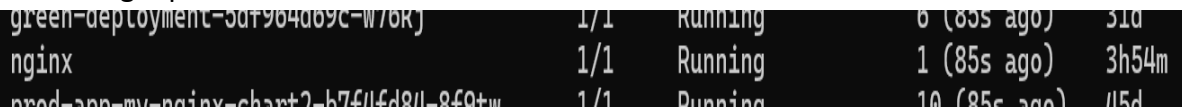
Cmd> kubectl apply -f pod.yaml



```
C:\Users\Dell\Documents\3-1 odd\ADAPTIVE SOFTWARE ENGINEERING\Lab
s>kubectl apply -f pod.yaml
pod/nginx created
```

Verify the Pod

> kubectl get pods



green-deployment-5d4964d69c-w76kj	1/1	Running	0 (85s ago)	31d
nginx	1/1	Running	1 (85s ago)	3h54m
pod/nginx	1/1	Running	10 (85s ago)	115d

Get Detailed Information

> kubectl describe pod nginx

```
C:\Users\Dell>kubectl describe pod nginx
Name:          nginx
Namespace:     default
Priority:       0
Service Account: default
Node:          minikube/192.168.49.2
Start Time:    Sat, 26 Sep 2026 10:04:01 +0530
Labels:        <none>
Annotations:   <none>
Status:        Running
IP:            10.244.1.2
IPs:           IP: 10.244.1.2
Containers:
  nginx:
    Container ID:  docker://58cb41949aa7b5abba44cb92fe85a6bc83f9d6e9477514fbc2315e96a73b2157
    Image:         nginx
    Image ID:      docker-pullable://nginx@sha256:abe47724e466aeb9a345d8e46a221c2fa8953c7848b4a3bd9976a7199f8cf2
    Port:         80/TCP
    Host Port:    80/TCP
    State:        Running
      Started:    Sat, 26 Sep 2026 13:57:58 +0530
    Last State:   Terminated
      Reason:     Error
      Exit Code:  255
    Started:      Sat, 26 Sep 2026 10:04:35 +0530
    Finished:     Sat, 26 Sep 2026 13:56:53 +0530
    Ready:        True
    Restart Count: 1
    Environment:  <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-85rr7 (ro)
Conditions:
  Type              Status
  PodReadyToStartContainers  True
  Initialized        True
  Ready              True
  ContainersReady    True
  PodScheduled       True
Volumes:
  kube-api-access-85rr7:
    Type:              Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds: 3607
    ConfigMapName:      kube-root-ca.crt
    Optional:           false
    DownwardAPI:        true
    BestEffort:         <none>
QoS Class:           BestEffort
Node-Selectors:      <none>
Tolerations:         node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
                     node.kubernetes.io/unreachable:NoExecute op=Exists for 300s
Events:
  Type    Reason          Age   From      Message
  ----    -
  Normal  SandboxChanged  5m23s  kubelet   Pod sandbox changed, it will be killed and re-created.
  Normal  Pulling         4m57s  kubelet   spec.containers[nginx]: Pulling image "nginx"
  Normal  Pulled          4m46s  kubelet   spec.containers[nginx]: Successfully pulled image "nginx" in 10.701s (10.701s including waiting). Image size: 162076942 bytes.
  Normal  Created         4m45s  kubelet   spec.containers[nginx]: Container created
  Normal  Started         4m45s  kubelet   spec.containers[nginx]: Container started
C:\Users\Dell>
```

Access the Pod

kubectl port-forward nginx 8080:80

Output

```
C:\Users\Dell\Documents\3-1 odd\ADAPTIVE SOFTWARE ENGINEERING\Lab
s>kubectl port-forward nginx 8080:80
Forwarding from 127.0.0.1:8080 -> 80
Forwarding from [::1]:8080 -> 80
```

Open a browser and visit: <http://localhost:8080>

